MIL-E-1/333B(EL)
8 November 1970
SUPERSEDING
MIL-E-1/333A
21 January 1959

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 2C51

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Twin triode, medium Mu

(B) Outline --- 6-1 (EIA)

Base --- E9-1

Envelope --- T6-1/2

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	h	1k	1g	1a	sh	2a	2g	2k	h

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ik/k	Rk/k	Pp/a	(B) Ehk	(B) Alt
Unit:	V	Vdc	Vdc	mAdc	Ohms	W	v	ft
Maximum:	6.9	330	---	20	---	1.6	100	(See note 1)
Minimum:	5.7	---	---	---	---	---	---	
TEST CONDITIONS:	6.3	150	0	---	240	---	---	---

GENERAL:

Qualification - Not required

Preproduction (See Note 9)

(B) denotes changes

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METH 3	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1236	Total grid current	Rk = 120 ohms (see notes 2, 3, and 4)	0.65	II	ΔIb	---	0.6	mAdc
1306	Transconductance (1)		0.65	II	Sm	4,500	6,500	μmhos
1306	Transconductance (2)	Ef = 5.7 V (see notes 2 and 5)	0.65	II	ΔSm Ef	---	15	%
(B) 1201	Short and discontinuity detection		0.4	II	---	---	---	---
1256	Electrode current (1) (anode)	See note 5	0.65	II	Ib	5.9	10.5	mAdc
1256	Electrode current (2) (anode)	Ec = -10 Vdc (see note 5 or 6)	0.65	II	Ib	---	75	μAdc
	Quality conformance inspection, part 2							
1301	Heater current		---	---	If	280	320	mA
(B) 1336	Heater-cathode leakage	See note 3	---	---	Ihk	---	30	μAdc
316	Amplification factor	See note 5	---	---	Mu	26	44	---
1331	Direct-interelectrode capacitance	Without shield (note 7)	---	---	Cgp	1.0	1.6	pF
		Without shield (note 7)			Cin	1.6	2.7	pF
		Without shield (note 7)			Cout	0.7	1.3	pF
		Without shield			Cpp	---	0.10	pF
(B) 1211	Insulation of electrodes		---	---	---	---	---	---
1031	Low-frequency vibration	Rk = 0; Ec = -3 Vdc; Rp = 2,000 ohms (see note 3)	---	---	Ep	---	150	mVac
(S) 1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
(B) 1506	Heater-cycling life	Ef = 7.5 Vdc; Ehk = 100 Vdc; Eb = Ec = 0; Rk = 0	---	---	---	---	---	---
---	Heater-cycling life-test end point	Heater-cathode leakage	---	---	Ihk	---	20	μAdc
---	Life-test provisions	See note 8	---	---	---	---	---	---
--	Life-test end points (500 hours)	Transconductance (1)	---	---	Sm	3,800	---	μmhos
		Total grid current	---	---	ΔIb	---	0.6	mAdc

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NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in basic document.
2. This test to be performed at the conclusion of the holding period.
3. Test with sections connected in parallel.
4. For this test, the change in anode current shall be measured when a resistance of 1.8 megohms is inserted in series with the grid.
5. Test each unit separately, but with test voltages applied to both units.
6. Test each unit separately, but with voltages given under "Test Conditions" applied to section not under this test.
7. Test each unit separately. Connect internal shield and heater to cathode of section under test.
8. Twenty tubes per week shall be tested. This number may be reduced to 10 tubes per week if there have been no failures for four consecutive weeks. In the event of failure, the manufacturer shall return to the more stringent number of 20 tubes per week.
9. When preproduction or First Article inspection is invoked, all tests listed herein shall be required. The sample size shall be six and the allowable defects zero for all tests other than the Life test where the conditions of Note 8 apply and the allowable defects shall not exceed one.

Custodian
Army-EL

Preparing activity:
Army-EL

Project Number 5960-A900

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